

Remember when Pluto was a planet? Until very recently, Pluto's planetary status was an immutable fact learned by primary school kids the world over.

Pluto had an undisputed place in papier-mâché models of the solar system and even its very own song lyric -- 'And last of all there's Pluto too...'

But in August 2006, Pluto found itself cruelly demoted to a dwarf planet after failing to meet one of three criteria the International Astronomical Union uses to define a full-sized planet.

We are taught from an early age to think of scientific facts as, well, facts. And it can be unsettling when this turns out to not be the case.

But does this uncertainty really mean that we can't trust science?

According to the Royal Society, the world's oldest independent scientific academy, the answer is quite the opposite.

Questioning established facts is actually at the heart of the scientific method.

Think of the whole process as a giant system of checks and balances based on raw information drawn from experiments

or observations of nature.

Those experiments and observations lead to the formulation of a hypothesis which then goes through a rigorous process of checks by other scientists.

However, there isn't always enough information to draw the right conclusions.

Take for example Sir Francis Bacon, who noticed that the coastlines of West Africa and eastern South America looked like they were symmetrical. He believed that nature was copying herself.

What he couldn't have known in the early 1620s was that the two coasts are two sides of a fault-line in an ancient supercontinent which split apart 140 million years ago.

Bacon's hypothesis remained gospel for centuries, until new discoveries about the science of plate tectonics emerged in the 1950s and provided a neat explanation for Bacon's observation.

There's also the fact that different experts examining the same raw data can draw different conclusions.

Spare a thought for poor Robert Plot who, in the 1670s, found a fossilised bone that he thought was that of a giant human.

At least he wasn't around when a geologist and zoologist

proved it was in fact a Megalosaurus, a kind of dinosaur that roamed the Earth in the middle Jurassic.

Sometimes the scientific method is all about setting up ground-breaking experiments at the right time, in the right place to test theories.

In 1919, a blockbuster meeting of the Royal Society confirmed Einstein's theory of general relativity after a solar eclipse provided the perfect circumstances to measure the bending of starlight.

This showed that the gravity of a massive object, such as the Sun, could bend light around it.

Under the watchful eye of Sir Isaac Newton's portrait, the scientific community replaced Newton's previous theory with Einstein's newer and more general interpretation of gravity.

But even with evidence in its favor, Einstein's theory still can't be thought of as a fundamental fact.

It's even possible that in the future new breakthroughs will supersede our understanding of general relativity.

Uncertainty in science isn't enough if all the perspectives are the same. Sometimes a lack of diversity in the system can skew the results.

Until the 1970s, primatologists were mostly men and tended

to limit their studies to male primates.

They believed that the aggression observed in baboons meant human evolution had been driven by similar behavior.

That was until anthropologist Shirley Strum began observing both female and male baboons, disproving these earlier theories and facing a backlash for her troubles.

Scientists don't always get it right on the first go. But rather than instilling doubt, it should inspire confidence that outdated ideas are replaced when new information becomes available.

It's the difference between upgrading your mobile and clinging to your old rotary phone because you don't want to be wrong.

And in most cases, newer breakthroughs would not be possible without the legwork that came before them.

While the rotary phone isn't the best available technology today, your smart phone wouldn't exist without it.

Uncertainty is baked into the scientific process. It's the fundamental reason that progress is possible.

Ultimately, it comes down to who you trust more, the person who's certain they're right, or the person who's willing to be proved wrong.